Analogue Electronic Circuits 2

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Analogue Electronic Circuits 2 (A05346)

Short Title:	Analogue Electronic Circuits 2		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
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Credits	5	**Level:**	Intermediate

Description of Module / Aims

The module provides the student with an essential grounding in the areas of Analogue circuit design. The module covers the fundamentals of circuit design and builds on the students' previous knowledge of ideal circuits. The minimisation and control of device induced errors is to the forefront in the teaching of this course.

Programmes

`ANEC-0002`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	2 / 4 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	2 / 4 / M

Indicative Content

- Small Signal Amplifiers, Review, Capacitor coupled stages, Direct coupled stages
- Negative Feed Back, Concepts, Voltage feedback, Current feedback, Stability, Bode Plot
- Linear Circuits, Op-Amp Three Stage Architecture, Deviations from ideal performance, Error analysis, Frequency Response, Compensated / Uncompensated
- Linear Circuits Design, High Gain amplifiers, Instrumentation amplifiers, Integrators, Differentiators, Specialist Op-Amp IC's, Practical circuits

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the fundamentals of analogue circuit design.
2. Explain the role of negative feedback in amplifier circuits.
3. Outline the sources of errors in amplifiers constructed from op-amps.
4. Develop linear op-amp circuit designs.
5. Analyse the frequency response and error response of different circuit designs.
6. Build, test, design specific elements of and verify the correct operation of circuits related to the theory based lectures.

Learning and Teaching Methods

- The course delivery will be by direct contact lecturing and supervised laboratory practical work. The supervised lab element will coordinate to the taught element and will be assessed through performance in the lab and the quality and punctuality of the students' lab reports.
- Problem sheets will be given to the student to assist in self-paced study.
- Tutorials will allow the student to build up their skills of analysing and solving problems
- In class assessment (Mid-term & End-term) allowed students to form the solid memory of their theory study, monitor their learning status and gain the experience with time limited examination

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4,5
Continuous Assessment	40%	
Practical	30%	6
In-Class Assessment	10%	1, 2, 3, 4, 5

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Bogart, T. _Electronic Devices and Circuits_. 6th Ed.. US: Prentice Hall, 2004.
- Malvino, A.B. and D. Bates. _Electronic Principles_. 8th ed. USA: McGraw Hill, 2015.

Supplementary Material

- Horowitz, P. and W. Hill. _The Art of Electronics_. 2nd Ed.. UK: Cambridge, 1989.

Requested Resources

- Lecture Room: Loose Seated
- ENGINEERING LAB: Electronics